

1. PROBLEM STATEMENT

- **High-Volume Screening:** Recruiters must rank large cohorts of resumes against one JD.
- **Keyword ATS Blindspots:** Misses synonyms, project context, and genuine experience depth.
- **Black-Box LLM Flaws:** Arbitrary 1–100 prompts are non-deterministic and un-auditable.
- **Requirement:** A deterministic, explainable dual matching and ranking pipeline.

2. SOLUTION OVERVIEW

SmartSelect ingests one Job Description and a batch of multi-format resumes, normalizes structured entities, evaluates candidates across dual **keyword token** and **semantic embedding** vectors, and computes deterministic convex combination scores. Every applicant receives an evidence-backed score, rank, and transparent shortlist explanation.

3. SYSTEM ARCHITECTURE & DATA FLOW



4. CORE TECH STACK

Backend: Python 3, FastAPI, Uvicorn

Frontend: Vanilla JS, CSS3 SPA

Parsers: pypdf, pdfplumber, python-docx, bs4

Keyword: RapidFuzz + Alias Normalization

Semantic: sentence-transformers (all-MiniLM-L6-v2)

Testing: pytest (34 Test Suites Passing)

5. WHY SMARTSELECT IS DIFFERENT

100% Deterministic & Auditable: No LLM hallucinations or guess scores. The engine evaluates verified project and role evidence, providing mathematical explanations for every shortlist decision.

6. CORE & DIFFERENTIATING FEATURES

Dual Keyword + Semantic Engine

Token matching and 384-dimensional dense vectors genuinely combine to determine candidate scores.

Top-3 & Candidate Profile Inspection

Surfaces matched/missing qualifications, score metrics, per-skill cards, and 1-to-1 semantic proof.

Skill Radar Cohort Analytics **BONUS**

Visualizes cohort qualification coverage and identifies applicant pool technical deficiency gaps.

Full Pool Deterministic Ranking

Ranks 100% of applicants from strongest to weakest with adjustable recruiter weight controls.

Rank Gap ("Why A over B?") **BONUS**

Head-to-head comparison tool isolating score deltas, keyword/semantic gaps, and key differentiators.

JD Inclusivity & Bias Audit **BONUS**

Audits JDs for exclusionary phrasing and overly restrictive criteria with actionable recommendations.

"SmartSelect transforms resume screening from a black-box scoring task into an explainable, evidence-driven shortlisting process."